

Experiment 13.1.1

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Algorithm –

1. Start
2. Input
3. Read max_value from the user.
4. Initialize
5. Create an empty list fib_sequence
6. Set a = 0 and b = 1
7. Loop Condition
8. While $a \leq \text{max_value}$, repeat steps 5–6:
9. Store Value
10. Append a to fib_sequence
11. Update Values
12. Set a = b
13. Set b = a + b (using previous values)
14. Exit Loop
15. When $a > \text{max_value}$, stop the loop
16. Output
17. Return (or display) fib_sequence
18. End

13.1.1: Fibonacci Sequence using Modules

Write a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value.

Module Requirements

- Create a module `fibonacci_module.py` with a function `generate_fibonacci_sequence(max_value)` that returns list of all Fibonacci numbers less than or equal to the maximum value.

Main Program Requirements

Create `main.py` that performs the following:

- Import the `fibonacci_module`.
- Read an integer `n` from user input.
- Validate if `n` is positive.
- If valid, call the module function and display the Fibonacci sequence.
- If invalid, display an error message.

Input Format:

- Single integer `n` representing the maximum value for Fibonacci sequence.

Output Format:

- If `n > 0`, print the Fibonacci sequence as space-separated numbers.
- If `n ≤ 0`, print: "Please enter a positive integer".

Note:

- Two files are provided: `fibonacci_module.py` and `main.py`. The `main.py` file handles input and output. You need to complete both files based on the question requirements.

Sample Test Cases

main.py

1import fibonacci_module
2
3n = int(input())
4
5if n > 0:
6 fib_series = fibonacci_module.generate_fibonacci_sequence(n)
7 print(' '.join(map(str, fib_series)))
8else:
9 print("Please enter a positive integer")

4 out of 4 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 0.004 s
Test case 2 0.019 s
Test case 3 19.00 ms

Expected output
0 1 1 2 3 5 8 +

Actual output
0 1 1 2 3 5 8 +

Flowchart –

